Software Requirements Specification (SRS)
IRPA Digital Board Governance System (IRPA-DBGS)
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Scope: Documentation only; no application, rules, production or deployment changes.
Evidence convention: IMPLEMENTED = directly evidenced in inspected source; GAP = requested or expected but not found/verifiable; PROPO
1. Introduction
1.1 Purpose
This SRS records the observable and source-level requirements of IRPA-DBGS so IRPA can review scope, roles, workflows, data, interfaces, c
1.2 Scope
IRPA-DBGS is a browser-first institutional governance workspace for IRPA board members, personnel, administrators, auditors and authorised
The source includes multiple execution boundaries: React/Vite frontend; Firebase Authentication and Firestore; Firebase Functions; Cloudflare
Out of scope: redesigning the business rules; asserting legal validity of electronic signatures; inferring donor/grant submission capability; asser
1.3 Stakeholders
- IRPA Board and Board Chairperson: governance, meetings, resolutions, voting and oversight.
- Executive Director and authorised approvers: institutional decisions, approvals and execution.
- Secretariat / administrator: identity lifecycle, invitations, meeting coordination and records.
- Department directors and personnel: departmental records, actions, reports and approvals.
- Finance and procurement personnel: financial controls, procurement and payment traceability.
- Communications personnel and approvers: editorial, website, social and media governance.
- Research/data personnel: collection, datasets, analysis, information exchange and release readiness.
- External auditors and special invitees: restricted read or meeting access.
- IT/system administrators: configuration, access and release diagnostics.
- Developers/maintainers: code, integrations, tests, deployment and recovery.
1.4 Definitions and abbreviations
- DBGS: Digital Board Governance System.
- Worker/Gateway: Cloudflare server-side API boundary; the Drive Gateway is named irpa-google-drive-gateway .
- Actual / Trial: data environments selected through the Data Environment Gate; identity/security collections are designated Actual-only in the d
- PWA: Progressive Web App.
- DO: Cloudflare Durable Object; ESIGN_DO is bound for e-signature records.
- KV: Cloudflare Workers KV; production gateway binding is DRIVE_KV .
- D1: Cloudflare SQL database used by the isolated parallel grant crawler.
- Lifecycle: controlled document state transitions, signature and archive stages.
- IMPLEMENTED / GAP / PROPOSED: evidence labels defined above.
1.5 Evidence and limitations
The source inventory and route registrations were inspected on the repository’s main baseline. The public live URL https://irpa-digital-board-go
2. Overall description
2.1 Product perspective / architecture
mermaid
flowchart TD
U[Browser / PWA / Android / Windows] -- FE[React 19 + Vite frontend]
FE -- FA[Firebase Authentication]
FE -- FS[Firestore + persistent local cache]
FE -- CF[Cloudflare frontend Worker / static assets]
FE -- GW[Google Drive Gateway Worker]
GW -- FS

GW -- GD[Google Drive OAuth + Drive files]
GW -- SMTP[SMTP mail.irpa.or.tz]
GW -- WP[WordPress staging content API]
GW -- CP[cPanel API / file operations]
GW -- SOC[Social provider OAuth APIs]
GW -- LK[LiveKit meeting media]
GW -- KV[Cloudflare KV]
GW -- DO[Cloudflare Durable Object: ESIGN_DO]
FE -- FN[Firebase Functions]
GC[Separate grant crawler Worker] -- D1[Isolated Cloudflare D1]
IMPLEMENTED architecture observations
- Frontend uses React 19, Vite and Firebase JS SDK 11; app navigation and role projection are primarily in src/App.jsx .
- Firebase config initializes Auth and Firestore with persistent local cache and multi-tab manager. Applicant enrollment has a separate named F
- The frontend Worker serves built assets from dist ; Firebase Hosting rewrites SPA routes to index.html .
- Drive Gateway uses Cloudflare Worker entry drive-gateway/src/index.js , router modules and Firestore REST/admin token logic. It is configure
- A separate grant crawler exists; its parallel implementation stores discovered public-call metadata in isolated D1 and explicitly does not yet sy
- Browser permission projection is not a security boundary; Firestore rules and trusted backend checks remain authoritative.
GAP: no single authoritative architecture diagram or complete runtime deployment inventory was found that proves all environments and bindi
2.2 Product functions
The product functions are captured by the functional requirements in 3 and page inventory in 6. The current NAV constant declares 31 top-lev
2.3 User classes and roles
Source-visible role families include Administrator; Executive Director; Board Chairperson, Vice Chairperson, Secretary, Treasurer and Board M
Role membership and effective access are resolved from active member/employee/auditor/admin profiles and permission policy. Departmental
2.4 Operating environment
- Web: modern browser with JavaScript, HTTPS, Firebase Auth and network access.
- Mobile: responsive web/PWA at 390 CSS px target width; manifest display mode standalone.
- Windows: Electron BrowserWindow; context isolation on, Node integration off, sandbox on; HTTPS URL with offline HTML fallback.
- Android: Capacitor; local dist webDir; no remote server.url to preserve local/offline startup.
- Server: Cloudflare Workers, Firebase Auth/Firestore/Functions, Google Drive, SMTP, WordPress and cPanel APIs; LiveKit for meeting media
- Build/test: Node.js/npm, Vite, Firebase emulator/rules test tooling, Playwright/browser tests and GitHub Actions workflows.
2.5 Design and implementation constraints
1. Cloudflare-first gateway: document references, controlled document lifecycle, signing, invitation mail and multiple governance services use C
2. Firebase plan and quotas: Firestore/Auth/Functions limits and pricing are external operational constraints. PROPOSED: document quota buc
3. Offline: service worker caches a limited static shell and excludes API/auth/signature/invitation/token/upload/approval/PDF traffic. Offline mod
4. Security: client-side navigation/permission projection is not authoritative; server and Firestore rules govern data.
5. Environment isolation: Actual-only identity/security records must not be routed into Trial data. Gateway staging and production are separate
6. No production mutation in this task: all work is documentation-only.
7. Integration credentials: secrets are environment-managed; they must never be printed, stored in source or returned by diagnostics.
2.6 Assumptions and dependencies
- Firebase project IDs, Cloudflare bindings, OAuth credentials, Google Drive permissions, SMTP mailbox and third-party accounts are provisior
- Network-dependent functions require network availability; static shell/offline fallback is the only evidenced offline guarantee.
- A valid, active institutional register record is required for protected identity and signing operations.
- WordPress editing is intentionally restricted to a dedicated staging WordPress host/account in the code-level website editor; cPanel productio
- GAP: no verified SLA, RPO/RTO, retention schedule, accessibility conformance statement or user-volume forecast was found.
3. Functional requirements
Priority meanings: P1 = governance/security critical; P2 = important operational capability; P3 = supporting capability. Acceptance criteria are te
Requirement ID Requirement Priority Acceptance criteria Portal / page
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GEN-FR-001 Authenticate users using the configured Firebase Authentication providers and preserve the authenticated session. P1 Login acc
GEN-FR-002 Resolve the user’s active institutional identity and role context from the member, employee, auditor or administrator registers. P1
GEN-FR-003 Display role-appropriate navigation and apply trusted Firestore/Worker authorization independently of client-side visibility. P1 Dire
GEN-FR-004 Provide the shared application shell, global navigation, in-portal navigation and page Back/Next navigation. P1 Navigation chang
GEN-FR-005 Require an explicit selection of Trial or Actual data environment before protected portal use; keep identity/security records in Actu
INV-FR-001 Allow an authorised operator to create and manage member/employee invitations linked to an institutional record. P1 Invitation sto
INV-FR-002 Send invitation messages through the configured gateway mail service and record delivery outcome. P1 A valid request returns a
INV-FR-003 Redeem an invitation token only for the matching authenticated email and valid invitation state. P1 Expired, cancelled, mismatched
INV-FR-004 Allow the invitee to create or configure a password and complete account activation. P1 Successful activation links the Firebase U
INV-FR-005 Support induction and registration records before institutional account activation. P2 An induction application is stored and lookup
MEM-FR-001 Maintain Board Member, Member and Employee institutional registers with unique references and status. P1 Create/update actio
MEM-FR-002 Associate one Firebase identity with both member and employee records where the person holds both capacities. P1 Activation
MEET-FR-001 Create, update and delete meeting register entries subject to role and lifecycle policy. P1 Only authorised actors can perform ea
MEET-FR-002 Issue, update and revoke meeting access invitations and authorise entry against meeting-specific policy. P1 A non-participant c
MEET-FR-003 Provide meeting room participation, activity and subscription views scoped to the current user and meeting. P1 A participant see
MEET-FR-004 Capture meeting records, proceedings, drafts, transcripts and translation outputs through gateway services. P2 Record retrieval
MEET-FR-005 Support meeting voting open/close, resolution, decision and action records. P1 Voting cannot be cast outside the open window
MEET-FR-006 Support meeting media tokens and recording start/stop/status through configured LiveKit credentials. P2 Missing LiveKit configu
DOC-FR-001 Issue a unique controlled-document reference through the Cloudflare service. P1 Reference issuance requires authentication and
DOC-FR-002 Upload controlled documents with type, classification, archive category, version and record-origin metadata. P1 Invalid file type/s
DOC-FR-003 List, retrieve, download and share documents according to document-level access policy. P1 Unauthorised users cannot list or d
DOC-FR-004 Apply document lifecycle transitions and retain lifecycle history. P1 Only permitted transitions succeed; each transition records ac
DOC-FR-005 Provision and retrieve department, signature and final archive locations in Google Drive. P1 Archive operations return verified ref
SIG-FR-001 Create a signature workflow/envelope and define ordered or optional signers and access-only recipients. P1 Recipients are validat
SIG-FR-002 Verify signer identity and authority against active member/employee registers before signature execution. P1 Manual authority text
SIG-FR-003 Capture signature input through the signature pad and bind signature fields to the document. P1 The signer confirms the intended
SIG-FR-004 Submit and complete signing through the gateway, preserving evidence and document hash. P1 Completion is atomic or remains
SIG-FR-005 Mirror completed signature state to Firestore and archive signed documents in Drive. P1 A failed mirror/archive does not display fa
SIG-FR-006 Support signer invitation, decline, void, status, evidence and completion retry operations. P1 Each action validates the token/actor
AUTH-FR-001 Create authorization requests with requester, reviewer, approver, implementer, basis, justification and linked evidence. P1 Requ
AUTH-FR-002 Enforce the Draft Submitted Under Review Approved/Rejected/Returned Completed state machine. P1 Only the designate
AUTH-FR-003 Record decision basis, conditions, authority confirmation, history and audit events. P1 Every transition writes immutable history
FIN-FR-001 Maintain finance budgets, transactions, funding, approvals, commitments, grants, accounts, reconciliations, assets, journals and re
FIN-FR-002 Maintain staff payment requests and payment trace references. P2 A trace has creator/updater identity and timestamp; finance ac
PROC-FR-001 Maintain procurement requests, vendors, scores, blacklist and probation status. P1 Vendor and procurement actions are permis
RISK-FR-001 Maintain risk register entries, reports, decisions and actions as governance evidence. P2 Restricted records are not exposed to u
RES-FR-001 Provide research studies, datasets, observations, analyses, knowledge, outputs, instruments and submissions. P2 Dataset record
RES-FR-002 Support information requests, information grants, shared information, updates, tool permissions and portal alerts. P2 A request/up
GRANT-FR-001 Discover public grant opportunities through crawler sources and persist opportunity metadata in its configured store. P2 Crawl
GRANT-FR-002 Support grant source registry, relevance screening and funding pipeline records. P2 Each source/opportunity retains provenan
COM-FR-001 Create and edit communications editorial items and website page/post drafts. P2 Only authorised authors can create/edit; revisio
COM-FR-002 Submit editorial content for approval and enforce separation of author and approver. P1 Author cannot approve own content; onl
COM-FR-003 Connect to WordPress staging for content status, list, read and controlled update operations. P2 Production WordPress host is re
COM-FR-004 Connect social accounts and use platform-specific OAuth/publishing routes where configured. P2 No publish is represented as su
COM-FR-005 Maintain communication/media asset metadata including URL, caption, credit and consent status. P2 Only HTTPS asset URLs a
SYS-FR-001 Expose administrator system diagnostics and release-readiness checks without returning secret values. P1 Only authorised admi
SYS-FR-002 Provide settings, administrator provisioning, department archive registry and department portal definitions. P2 Only designated ro
SYS-FR-003 Provide downloadable application resources and release artefacts. P3 Only existing downloadable artefacts are listed; missing ar

OFF-FR-001 Install as a PWA with a versioned service-worker shell and manifest. P1 Install metadata and shell assets load; API/auth/signature
OFF-FR-002 Provide an offline fallback for navigation shell startup and static assets. P1 When network is unavailable, cached shell or offline pa
OFF-FR-003 Package Windows desktop using Electron with context isolation, sandbox and no Node integration. P2 Desktop app loads configu
OFF-FR-004 Package Android with Capacitor and local dist web assets for offline startup. P2 Android config uses local webDir and does not fo
REL-FR-001 Run automated unit, policy, integration and browser regression checks before release. P1 Release evidence records command, c
REL-FR-002 Deploy frontend and gateway through separate controlled workflows with environment isolation. P1 Production deployment requir
REL-FR-003 Maintain documented rollback path for frontend, Worker and data migration changes. P1 Rollback identifies immutable release re
4. External interface requirements
4.1 User interfaces
The shared shell includes branded header/identity area, global portal category navigation (IT, Governance, Finance, Evidence, Funding, Comm
4.2 Software interfaces
- Firebase Authentication: email/password, Google sign-in/redirect, magic link and invitation activation/password setup paths.
- Firestore: institutional registers, meeting/governance, finance/procurement, research, grant and communications records; client uses persiste
- Cloudflare Drive Gateway: REST routes in 4.4; Firebase ID token authentication and server-side Firestore admin access.
- Google Drive: OAuth start/callback, folder provisioning, controlled document storage, archive and download.
- SMTP: gateway mail delivery; configured host in Wrangler is mail.irpa.or.tz , port 465, sender info@irpa.or.tz (configuration values, not creder
- WordPress: content/status integration via communicationsSite.mjs ; editor restricted to staging host and dedicated Editor account.
- cPanel: account-level API/file operations constrained by environment and document root.
- Social platforms: Meta, LinkedIn, X, YouTube and WhatsApp configuration hooks; actual connected status is not proven by source.
- LiveKit: meeting media token/recording services; runtime requires LIVEKIT_URL/API key/secret.
- Grant discovery: separate Cloudflare Worker and D1 database; parallel crawler is isolated and Firebase sync intentionally disabled.
4.3 Communications integration configuration
Code reads WordPress WORDPRESS_BASE_URL , WORDPRESS_USERNAME , WORDPRESS_APPLICATION_PASSWORD ; cPanel CP
4.4 Gateway route inventory
The following route groups are present in the inspected Worker dispatch and delegated routers. Braced alternatives denote route families; meth
- GET /health
- POST /oauth/start
- GET /oauth/callback
- POST /api/upload
- POST /api/upload-controlled-document
- POST /api/upload-controlled-document/policy-check
- POST /api/upload-controlled-document/register
- POST /api/download
- POST /api/delete
- POST /api/documents/next-reference
- GET POST /api/document-lifecycle/{list,get,upload,download,transition,signature-complete,share}
- POST /api/signature-pathway/reconcile
- GET /api/signature-workflow/list
- GET /api/signature-workflow/view
- POST /api/signature-workflow/add-signer-dry-run
- POST /api/signature-workflow/draft
- POST /api/signature/authority
- POST /api/signature-profile/folder
- POST /api/signature-profile/finalize
- POST /api/document-archive/folder
- POST /api/document-archive/provision
- POST /api/meeting-archive/provision
- POST /api/meeting-archive/get

- POST /api/signed-document/archive
- POST /api/signature-workflow/folder
- POST /api/signature-file/download
- GET POST /api/system/diagnostics
- GET POST /api/system/actual-data-access
- POST /api/system/release-readiness
- POST /api/members/reconcile-registered-identity-uids
- POST /api/invitations/redeem
- POST /api/invitations/session-state
- POST /api/invitations/password-set
- POST /api/send-invitation-email
- POST /api/invitations/send
- POST /api/signature-invitations/send
- POST /api/induction/lookup
- POST /api/session/profile
- POST /api/communication-media-archive/provision
- POST /api/research-archive/provision
- POST /api/meeting-register/{create,update,delete}
- POST /api/meeting-voting/{open,close}
- POST /api/meeting-decisions/create
- POST /api/meeting-participants/{update,delete}
- POST /api/meeting-invitations/update
- POST /api/meeting-room/activity
- POST /api/meeting-subscriptions/mine
- POST /api/meeting-records/{capture,draft,list,retrieve,protection,dispose}
- POST /api/meeting-proceedings/save
- POST /api/meeting-transcripts/translate
- POST /api/meeting-reports/compile-email
- POST /api/meeting-media/{token,recording/start,recording/stop,recording/status}
- GET POST /api/communications/publishing/pause
- GET POST PATCH /api/communications/editorial/{items,audit,items/{id},items/{id}/submit,items/{id}/approve,items/{id}/reject}
- GET /api/communications/site/status
- GET /api/communications/website/content
- PUT /api/communications/website/content
- GET POST /api/communications/social/
- POST /api/esign/cleanup
- POST /api/documents
- GET /api/documents/{id}/pdf
- POST /api/envelopes
- GET /api/envelopes/{id}[/status]
- POST /api/envelopes/{id}/{void,retry,retry-completion,resync}
- GET /api/sign/{open,status,evidence,pdf}
- POST /api/sign/{verify,submit,decline}

Route coverage note: dynamic delegated routes are represented by their route families. Exact per-route authorization, payload schemas and re

4.5 Error handling

The gateway returns JSON errors with HTTP status for authentication, authorization, validation, not-found, conflict, unavailable and internal fail

5. Non-functional requirements

NFR ID Quality attribute Requirement / verification

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NFR-SEC-001 Authentication Protected actions require valid Firebase identity; verify token expiry and invalid-token handling.
NFR-SEC-002 Authorization Enforce least privilege at Firestore/Worker; direct API tests must deny unauthorised access even when UI is bypa
NFR-SEC-003 Separation of duties Requester cannot approve own authorization; communications author cannot approve own item; signer role
NFR-SEC-004 Data isolation Trial and Actual records remain separated; Actual-only collections include identity/security registers, audit, signatu
NFR-SEC-005 Secret handling No secrets in client bundle, source control, logs or diagnostic responses; verify redaction and secret presence c
NFR-AUD-001 Auditability Governance and finance mutations record actor, action, record reference and timestamp; verify event correlation an
NFR-REL-001 Reliability Signature completion/mirroring and document archive operations must fail visibly and support controlled reconciliation
NFR-PERF-001 Performance PROPOSED: define measurable page/API latency budgets and test at representative dataset sizes; no numeric
NFR-ACC-001 Accessibility PROPOSED: WCAG 2.2 AA target; keyboard navigation, focus visibility, labels, contrast, zoom and screen-reader
NFR-RESP-001 Responsive UI PROPOSED target: support 390 844 mobile and 1440 900 desktop without horizontal page overflow or control
NFR-OFF-001 Offline Cache static shell only; do not cache authentication, signatures, uploads, approvals or API traffic. Display clear offline sta
NFR-BCP-001 Backup/recovery GAP: no complete tested recovery plan, RPO/RTO, Firestore export schedule, Drive recovery procedure or ke
NFR-MAINT-001 Maintainability Keep Worker route contracts, collection catalogue, page inventory and requirement traceability current with co
NFR-PRIV-001 Privacy Restrict member, employee, auditor, meeting, signature and finance records by need-to-know; avoid exposing private i
6. Data model
6.1 Firestore collection inventory
The COLLECTIONS registry in src/firebase/data.js explicitly names: members , employees , employeeCounters , memberCounters , participan
6.2 Key fields and relationships
- members / employees: institutional person identity, role(s), email, status, registration/activation status, Firebase UID, invitation linkage, depart
- invitations: recipient email/name, role/member type, institutional record type/id, member/employee/board-member linkage, status, accepted U
- meetings: meeting metadata, participants, access/subscription records, agenda/proceedings, meeting records, media/recording status and rel
- documents: reference, title, type, archive category, classification, version, origin, Drive identifiers, lifecycle stage, parent-document and signat
- signatureProfiles / signatureEnvelopes / signatures / signatureEvents: signer profile, envelope/document linkage, recipients/order, state, field
- workflowActions / authorizationRequests / audit: request fields, participants (UIDs), status, stage index/version, decision record, history, relate
- finance and procurement : finance-specific registers and ledger structures; source identifies named collections but a full canonical field schem
- research : study/dataset/observation/analysis/knowledge/output/instrument/submission and information exchange records.
- communications: editorial items/audit are stored in Cloudflare KV; media/social/website data also use Firestore or remote APIs as implemente
- grant opportunities: parallel discovery worker uses isolated D1 tables/migrations; not the same persistence boundary as the Firestore donor/fu
6.3 Relationship diagram
mermaid
erDiagram
MEMBERS --o{ INVITATIONS : receives
EMPLOYEES --o{ INVITATIONS : receives
MEMBERS }o--o EMPLOYEES : same_firebase_uid
MEETINGS --o{ PARTICIPANTS : has
MEETINGS --o{ RESOLUTIONS : produces
MEETINGS --o{ VOTES : collects
MEETINGS --o{ ACTIONS : tracks
DOCUMENTS --o{ SIGNATURE_ENVELOPES : routed_to
SIGNATURE_ENVELOPES --o{ SIGNATURES : captures
SIGNATURE_ENVELOPES --o{ SIGNATURE_EVENTS : records
DOCUMENTS --o{ WORKFLOW_ACTIONS : supports
WORKFLOW_ACTIONS --o{ AUDIT : audited_by
RESEARCH_STUDIES --o{ RESEARCH_DATASETS : contains
RESEARCH_DATASETS --o{ RESEARCH_OBSERVATIONS : contains
Relationship lines are logical links inferred from explicit source references; exact Firestore document paths and field-level cardinalities are not g
7. Workflow descriptions

7.1 Appointment acceptance registration invitation password activation
1. Administrator/authorised operator creates or identifies the institutional member/employee record.
2. Invitation record links to the canonical institutional record and carries role/member type and recipient email.
3. Gateway sends the invitation email through configured SMTP.
4. Invitee opens the invitation link and authenticates using the invitation email.
5. Gateway validates token/session state, expiry, cancellation and email binding.
6. Invitee sets/configures password; activation links Firebase UID to member/employee record(s).
7. Invitation status becomes Activated; protected navigation is resolved from active profiles.
8. GAP: the complete human appointment/board resolution/acceptance process before the register entry is not fully evidenced as one end-to-e
7.2 Upload signature approval archive
1. User selects Trial/Actual environment and requests controlled document reference.
2. User uploads with type/classification/version/archive metadata; gateway policy checks and registers document.
3. Document is assigned to signature workflow; signer order, authority and signature fields are validated.
4. Signers receive invitations and sign through authenticated signer UI/pad or controlled token flow.
5. Gateway records signature evidence/hash and completion state; mirror into Firestore.
6. Document lifecycle transitions to post-signature and final archive; Drive archive reference is recorded.
7. Reconciliation/retry handles incomplete mirror/archive; completion is not shown until verified.
8. Authorisation/approval workflows are distinct from e-signature and must be linked explicitly where required; signature alone does not equal i
7.3 Meetings
1. Authorised user creates meeting register entry and participants.
2. Meeting access invitation issued; gateway verifies identity, participant membership and meeting scope.
3. Meeting Room supports activity, subscriptions, working documents, attendance, media/recording and proceedings where configured.
4. Resolutions, votes, decisions and actions are recorded under meeting policy.
5. Reports/transcripts may be compiled, translated and emailed.
6. Retention/disposal is subject to protection/legal hold and server policy.
7.4 Grant intelligence
1. Discovery Worker polls public RSS/Atom and official donor sources (parallel crawler also has optional Brave Search).
2. Candidates are normalized, deduplicated and screened for relevance, deadline, geography and eligibility.
3. Parallel Worker writes metadata to isolated D1; production Firebase synchronization is intentionally disabled in that deployment.
4. Frontend Grant Intelligence / source registry supports source/opportunity review.
5. GAP: source discovery is not equivalent to a complete donor application drafting/submission workflow; no auto-submission is asserted.
7.5 Communications and media
1. Author creates editorial post or website-page draft with language, platform, link/image and optional schedule.
2. Author submits; approver reviews and approves or returns/rejects; author cannot self-approve.
3. Global publishing pause is administrator-controlled.
4. WordPress Website Studio reads staging pages/posts and updates allowed fields only; cPanel integration handles website file operations un
5. Media Library stores metadata (HTTPS asset URL, caption, credit, consent status); asset upload is separate.
6. Social account OAuth/status/publishing routes operate only when provider credentials and connection state are configured.
7. GAP: provider-by-provider production publishing, webhook delivery and post-publication reconciliation were not live-tested.
8. Verification, release and rollback
8.1 Test strategy
- Unit/policy tests: authorization, invitation, meeting categories, finance, communications, signature policy, document lifecycle.
- Rules tests: Firestore security rules using emulator/rules-unit-testing.
- Gateway tests: route contracts, authentication, upload lifecycle, Drive ID reconciliation, signature atomic completion, communications, release
- Browser tests: authenticated governance regression, portal navigation, meeting portal, fundraising registry responsiveness.
- PWA tests: service worker install/offline shell, excluded request classes.
- Build tests: root Vite build; desktop package build; Android APK/release workflows.
- Integration tests: staging only, isolated WordPress/cPanel and social/OAuth credentials; never run production mutations for routine verificatio

8.2 Release gates
1. Code review and CI pass against immutable commit.
2. Staging Worker/frontend build and integration tests pass with isolated staging resources.
3. Firestore rules tests and authorization boundary tests pass.
4. Signature lifecycle acceptance validates create sign mirror archive, including failure/retry.
5. PWA/Android/Windows package smoke tests pass.
6. Release Readiness diagnostics show no unresolved critical failures and do not expose secrets.
7. Explicit human production authorization; no implicit production deployment from documentation PR.
8.3 Rollback
- Frontend: redeploy the last known-good immutable build/commit through the approved workflow.
- Worker: restore the previous Worker version/configuration via controlled deployment history; verify bindings and route health.
- Firestore: restore from verified export/backup or perform a reviewed compensating correction; do not assume code rollback reverses data wri
- Google Drive: recover from retained versions/archives and reconcile Firestore references; do not delete evidence during rollback.
- WordPress/cPanel: restore prior content/file backup using staging validation first.
- Android/Windows: publish prior signed package and document version compatibility.
- GAP: verified RPO/RTO, backup cadence, restore drill evidence and a named rollback owner were not located.
9. Known gaps, defects and open decisions
9.1 Confirmed source-level gaps / risks
- GAP-001 Gateway disconnected in app shell: IRPA_GATEWAY_CONNECTED=false is explicitly set in src/App.jsx ; comments describe a te
- GAP-002 Temporary data gate: IRPA_TEMPORARY_DATA_GATE_ENABLED=true ; operational implications and removal criteria are undo
- GAP-003 Live runtime unverified: deployed web app could not be inspected with an authenticated session; live role/menu, integration reacha
- GAP-004 Page count discrepancy: requested baseline said 26 pages; source NAV currently contains 31 top-level modules and 15 nested su
- GAP-005 Offline scope: service worker caches app shell/assets only and excludes auth, APIs, signature, invitation, upload, approval and PD
- GAP-006 Collection schema: collection names are discoverable, but there is no single complete typed field schema and relationship contrac
- GAP-007 Backup/DR: no end-to-end tested backup/restore, RPO/RTO or recovery evidence was established.
- GAP-008 Performance/accessibility: no numeric latency SLO or WCAG conformance test report verified.
- GAP-009 Grant crawler integration: parallel crawler D1 is isolated and Firebase sync is intentionally off; do not assume its discovered record
- GAP-010 Social/website integrations: WordPress/cPanel/social code paths exist, but live credentials, account permissions and actual produc
- GAP-011 Android/Windows distribution: packaging workflows/configuration exist; current signed artifacts, distribution channel, support matrix
- GAP-012 Approval vs signature: institutional approval workflow and e-signature workflow are separate; policy for mandatory ordering/linkage
9.2 Decisions requiring IRPA input
1. Is the approved scope 26 pages, or the 31 NAV modules + 15 sub-portals currently in source?
2. What is the canonical role catalogue and who approves changes to role-to-portal and role-to-action permissions?
3. Which workflow is authoritative for appointment/board acceptance before invitations are issued?
4. Which document categories require approval, signature, both, or neither, and in what order?
5. What are the approved retention periods, legal-hold triggers, disposal authorities and archive ownership?
6. What offline actions, if any, may be drafted locally and synchronized later? Current code does not evidence offline writes for protected workfl
7. What backup frequency, RPO/RTO, recovery owner and annual restore test are required?
8. Is WordPress editing strictly staging-only, and what is the separately approved production publishing process?
9. Which social platforms are approved for live publishing and which roles may authorize posts?
10. Should grant-crawler D1 results remain isolated, or is a reviewed least-privilege sync to Firestore required?
11. What target browsers, Android versions, Windows versions and accessibility conformance level should be supported?
12. What are numeric performance targets and expected concurrent user/data volumes?
10. Traceability and page inventory
10.1 Top-level portal/page inventory
The source NAV contains 31 top-level modules.
Page ID Page / module Navigation group Source
--- --- --- ---

P-IT-OPERATIONS IT Operations IT Top-level NAV registry
P-DASHBOARD Dashboard GOVERNANCE Top-level NAV registry
P-INDUCTION-AND-ORIENTATION Induction and Orientation GOVERNANCE Top-level NAV registry
P-INDUCTION-APPLICATIONS Induction Applications GOVERNANCE Top-level NAV registry
P-BOARD-MEMBERS-REGISTRATION Board Members Registration GOVERNANCE Top-level NAV registry
P-MEMBERS-AND-PERSONNEL Members & Personnel GOVERNANCE Top-level NAV registry
P-INVITATIONS Invitations GOVERNANCE Top-level NAV registry
P-MEETINGS Meetings GOVERNANCE Top-level NAV registry
P-RESOLUTIONS Resolutions GOVERNANCE Top-level NAV registry
P-VOTING Voting GOVERNANCE Top-level NAV registry
P-ACTIONS Actions GOVERNANCE Top-level NAV registry
P-DOCUMENTS Documents GOVERNANCE Top-level NAV registry
P-SIGNATURE-PLATFORM Signature Platform GOVERNANCE Top-level NAV registry
P-DECISIONS Decisions GOVERNANCE Top-level NAV registry
P-RISK-REGISTER Risk Register GOVERNANCE Top-level NAV registry
P-AUTHORIZATION-AND-APPROVALS Authorization & Approvals GOVERNANCE Top-level NAV registry
P-EMPLOYEE-PAYMENTS Employee Payments GOVERNANCE Top-level NAV registry
P-RESEARCH-STATISTICS-AND-KNOWLEDGE Research, Statistics and Knowledge GOVERNANCE Top-level NAV registry
P-FINANCE-PORTFOLIO Finance Portfolio FINANCE Top-level NAV registry
P-PROCUREMENT Procurement FINANCE Top-level NAV registry
P-REPORTS Reports EVIDENCE Top-level NAV registry
P-AUDIT-TRAIL Audit Trail EVIDENCE Top-level NAV registry
P-GRANT-INTELLIGENCE-AND-FUNDING-OPPORTUNITIES Grant Intelligence & Funding Opportunities FUNDING Top-level NAV registry
P-COMMUNICATION-AND-MEDIA Communication & Media COMMUNICATIONS Top-level NAV registry
P-DOWNLOADS Downloads SYSTEM Top-level NAV registry
P-SETTINGS Settings SYSTEM Top-level NAV registry
P-ADD-ADMINISTRATOR Add Administrator SYSTEM Top-level NAV registry
P-DEPARTMENT-AND-ARCHIVE-REGISTRY Department and Archive Registry SYSTEM Top-level NAV registry
P-DEPARTMENT-PORTAL-BUILDER Department Portal Builder SYSTEM Top-level NAV registry
P-EXTERNAL-AUDITOR-PORTAL External Auditor Portal SYSTEM Top-level NAV registry
P-AUDITORS-AND-SPECIAL-INVITEES Auditors & Special Invitees SYSTEM Top-level NAV registry

10.2 Nested sub-portals

The source registers 15 nested sub-portals in PORTAL_ROLLER_GROUPS .

Page ID Sub-portal Parent Source
--- --- --- ---
SP-MEETINGS-MEETING-ROOM Meeting Room Meetings PORTAL_ROLLER_GROUPS
SP-MEETINGS-PARTICIPANTS Participants Meetings PORTAL_ROLLER_GROUPS
SP-MEETINGS-RESOLUTIONS Resolutions Meetings PORTAL_ROLLER_GROUPS
SP-MEETINGS-VOTING Voting Meetings PORTAL_ROLLER_GROUPS
SP-MEETINGS-ACTIONS Actions Meetings PORTAL_ROLLER_GROUPS
SP-MEETINGS-DECISIONS Decisions Meetings PORTAL_ROLLER_GROUPS
SP-DOCUMENTS-MY-DOCUMENTS My Documents Documents PORTAL_ROLLER_GROUPS
SP-DOCUMENTS-DEPARTMENT-DOCUMENTS Department Documents Documents PORTAL_ROLLER_GROUPS
SP-DOCUMENTS-SHARED-WITH-ME Shared With Me Documents PORTAL_ROLLER_GROUPS
SP-DOCUMENTS-SIGNING-WORKSPACE Signing Workspace Documents PORTAL_ROLLER_GROUPS
SP-DOCUMENTS-POST-SIGNATURE-WORK Post-Signature Work Documents PORTAL_ROLLER_GROUPS
SP-DOCUMENTS-FINAL-ARCHIVE Final Archive Documents PORTAL_ROLLER_GROUPS
SP-DOCUMENTS-UPLOAD-DOCUMENT Upload Document Documents PORTAL_ROLLER_GROUPS

SP-DOCUMENTS-RETRIEVE-ARCHIVED-DOCUMENTS Retrieve Archived Documents Documents PORTAL_ROLLER_GROUPS
SP-INDUCTION-AND-ORIENTATION-INDUCTION-APPLICATIONS Induction Applications Induction and Orientation PORTAL_ROLLER_GRO
10.3 Requirement page test traceability
Each functional requirement row in 3 identifies its page/portal. The tests below are source-discovered test candidates; where no direct test file
Requirement family Page mapping Verification source / test
--- --- ---
GEN Shared shell, all pages tests/portal-workspace-navigation.test.mjs ; tests/portal-access-blockers.test.mjs
INV / MEM Invitations, Activation, Members, Employees tests/invitation-removal-policy.test.mjs ; drive-gateway/tests/invitation-redemption.test.
MEET Meetings, Meeting Room, Participants, Resolutions, Voting, Actions tests/meeting-category-policy.test.mjs ; tests/meeting-portal-browse
DOC Documents and archive sub-portals tests/document-upload-archive-diagnostics.test.mjs ; drive-gateway/tests/document-lifecycle-router.te
SIG Signature Platform / signing sub-portals tests/signature-pathway-contract.test.mjs ; drive-gateway/tests/signature-complete-atomic-commit
AUTH Authorization & Approvals tests/authorization-functions-integration-gate.yml workflow; PROPOSED: authorization transition unit/browser
FIN / PROC Finance Portfolio, Employee Payments, Procurement tests/finance-portal-policy.test.cjs ; PROPOSED: ledger reconciliation integra
RES Research, Statistics and Knowledge tests/research-portal-security.test.mjs ; PROPOSED: dataset registry and release-readiness browser
GRANT Grant Intelligence tests/fundraising-source-registry.test.mjs ; cloudflare/grant-crawler-parallel/test/worker.test.js
COM Communication & Media tests/communication-media-policy.test.mjs ; drive-gateway/tests/communications-routes.test.mjs ; drive-gateway
OFF PWA, Android, Windows tests/pwa-install-offline.test.mjs ; tests/offline-build-workflow-policy.test.mjs ; Android/desktop GitHub workflows
REL IT Operations / Release Readiness tests/release-readiness-entry.test.mjs ; drive-gateway/tests/release-readiness-routes.test.mjs
Appendix A Source files inspected
- src/App.jsx , src/main.jsx , src/firebase/config.js , src/firebase/data.js , src/firebase/auth.js
- src/firebase/invitationWorkflow.js , src/firebase/authorizationWorkflow.js , src/firebase/documentLifecycle.js , src/firebase/signaturePlatform.js
- drive-gateway/src/index.js , router.mjs , communicationsRoutes.mjs , releaseReadinessRoutes.mjs , communicationsSite.mjs , meetingServic
- drive-gateway/wrangler.toml , wrangler.jsonc , firebase.json , public/sw.js , public/manifest.webmanifest , electron/main.cjs , desktop/package
- tests/ , drive-gateway/tests/ , .github/workflows/ , cloudflare/grant-crawler-parallel/
Appendix B Status disclaimer
This is a reverse-engineered documentation baseline from repository source at the stated commit. It is not a security certification, legal opinion,
Appendix C Additional delegated route registrations (expanded route coverage)
The gateway also delegates the following exact route groups. These are included here to avoid collapsing independently dispatched interfaces
C.1 WordPress and cPanel website gateway
- GET /api/communications/site/status
- GET /api/communications/website/content and GET /api/communications/website/content/{type}/{id}
- PUT /api/communications/website/content
- POST /api/communications/website/publish
- GET /api/communications/website/revisions
- POST /api/communications/website/restore
- GET /api/communications/website/media
- GET POST PUT DELETE /api/communications/cpanel/files (method availability is route-handler dependent)
- POST /api/communications/cpanel/rollback
- GET POST /api/communications/records
C.2 Grant crawler Worker (separate deployment)
- GET /health
- GET /opportunities (Firebase ID token)
- POST /crawl (control token)
- POST /run (Firebase ID token)
- POST /assistant/analyze (control token or Firebase ID token)
- POST /fundraising/search (control token or Firebase ID token)
- GET / , GET /application , GET /application/drafts , GET /application/drafts/{id} and delegated application-portal methods (application portal ha
- Scheduled Worker trigger scheduled() (platform event, not an HTTP route)

C.3 Frontend static-asset Worker

- GET /__irpa_health static frontend service health metadata; all other requests pass to the ASSETS binding.

C.4 Route inventory qualification

The Drive Gateway also dispatches the e-signature sub-router for /api/documents , /api/envelopes , and /api/sign/ when ESIGN_MODULE_EN

Appendix D Source page-component to wireframe traceability

The repository contains 41 JSX page components under src/pages/ (including the fundraising subfolder). Components that share a top-level pc

Source page component	Wireframe page ID
src/pages/AddAdministratorPortal.jsx	P-ADD-ADMINISTRATOR
src/pages/AuditorsSpecialInvitees.jsx	P-AUDITORS-AND-SPECIAL-INVITEES
src/pages/AuthorizationApprovals.jsx	P-AUTHORIZATION-AND-APPROVALS
src/pages/AuthorizationWorkflow.jsx	M-AUTHORIZATION-WORKFLOW
src/pages/BoardMembers.jsx	P-BOARD-MEMBERS-REGISTRATION
src/pages/CommunicationMediaOperations.jsx	M-COMMUNICATION-MEDIA-OPERATIONS
src/pages/CommunicationMediaPortal.jsx	P-COMMUNICATION-AND-MEDIA
src/pages/CommunicationMediaStudio.jsx	M-COMMUNICATION-MEDIA-STUDIO
src/pages/CommunicationsConnectionsFunding.jsx	M-COMMUNICATIONS-CONNECTIONS-FUNDING
src/pages/CommunicationsSocialStudio.jsx	M-COMMUNICATIONS-SOCIAL-STUDIO
src/pages/CommunicationsWebsiteStudio.jsx	M-COMMUNICATIONS-WEBSITE-STUDIO
src/pages/CredentialInterview.jsx	M-CREDENTIAL-INTERVIEW
src/pages/DocumentLifecycleModule.jsx	P-DOCUMENTS
src/pages/EmployeePayments.jsx	P-EMPLOYEE-PAYMENTS
src/pages/Employees.jsx	P-MEMBERS-AND-PERSONNEL
src/pages/ExternalAuditorPortal.jsx	P-EXTERNAL-AUDITOR-PORTAL
src/pages/FinancePortfolio.jsx	P-FINANCE-PORTFOLIO
src/pages/GatewaySigningPortal.jsx	M-GATEWAY-SIGNING-PORTAL
src/pages/GovernanceWorkspaces.jsx	P-DASHBOARD
src/pages/GrantIntelligencePortal.jsx	P-GRANT-INTELLIGENCE-AND-FUNDING-OPPORTUNITIES
src/pages/ITOperationsPortal.jsx	P-IT-OPERATIONS
src/pages/InductionAdmin.jsx	P-INDUCTION-APPLICATIONS
src/pages/InductionOrientation.jsx	P-INDUCTION-AND-ORIENTATION
src/pages/InvitationActivation.jsx	M-INVITATION-ACTIVATION
src/pages/Invitations.jsx	P-INVITATIONS
src/pages/MeetingEntryGateway.jsx	M-MEETING-ENTRY-GATEWAY
src/pages/MeetingRoom.jsx	SP-MEETINGS-MEETING-ROOM
src/pages/Meetings.jsx	P-MEETINGS
src/pages/Members.jsx	P-MEMBERS-AND-PERSONNEL
src/pages/OperationalGateways.jsx	M-OPERATIONAL-GATEWAYS
src/pages/OperationalGatewaysParticipants.jsx	M-OPERATIONAL-GATEWAYS-PARTICIPANTS
src/pages/ProcurementPortal.jsx	P-PROCUREMENT
src/pages/ResearchStatisticsKnowledgePortal.jsx	P-RESEARCH-STATISTICS-AND-KNOWLEDGE
src/pages/Resolutions.jsx	P-RESOLUTIONS
src/pages/Settings.jsx	P-SETTINGS
src/pages/SignaturePlatform.jsx	P-SIGNATURE-PLATFORM
src/pages/SignaturePlatformWithUpload.jsx	P-SIGNATURE-PLATFORM
src/pages/SocialMediaAccounts.jsx	M-SOCIAL-MEDIA-ACCOUNTS
src/pages/Voting.jsx	P-VOTING
src/pages/WorkflowEngines.jsx	M-WORKFLOW-ENGINES

src/pages/grant-fundraising/FundraisingSourceRegistry.jsx M-FUNDRAISING-SOURCE-REGISTRY

D.1 Additional subsystem component screens

The following component-level capabilities are represented by dedicated wireframes in addition to the 41 source page components:

- M-RESEARCH-DATA-COLLECTION Research Data Collection Studio.
- M-DATASET-REGISTRY Dataset Registry (subsystem screen; distinct page identity is not a top-level NAV item).
- M-RELEASE-READINESS Release Readiness component.
- M-SIGNATURE-PATHWAY-SELF-TEST Signature Pathway Self-Test.
- M-STATISTICAL-ANALYSIS-WORKBENCH Statistical Analysis Workbench.
- M-RESEARCH-EVIDENCE-EXCHANGE Research Evidence Exchange.
- X-SIGNATURE-PAD Signature field confirmation pad.
- X-UPLOAD-FLOW Controlled upload workflow.
- X-INVITATION-FLOW Invitation and account activation workflow.

D.2 Wireframe package counts

- 31 top-level NAV modules.
- 15 registered nested sub-portals.
- 21 additional source-component/subsystem wireframes.
- 5 shared shell/authentication/workflow wireframes.
- 72 HTML wireframes , plus this index; one PDF bundle contains 72 page-specific wireframe summaries.